

# Statistics

Hypothesis Testing

# Hypothesis Testing



## What Is Hypothesis Testing?

Hypothesis testing is a way for scientists or researchers to **test ideas or claims** using data.

Think of it like a **trial**: you make a claim (the hypothesis), collect evidence (data), and then decide if your claim makes sense based on the evidence.

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## Real-Life Example

Let's say school canteen says:

"Students eat **an average of 2 apples a day.**"

You and your friends think, "That can't be right. We don't eat that many apples!"

So you decide to **test their claim** by collecting data.

## Important Terms

- **Hypothesis:** A guess or claim you test.
  - **Null Hypothesis ( $H_0$ ):** The claim you're testing (usually that nothing has changed).
  - **Alternative Hypothesis ( $H_1$ ):** The opposite of the null (something **has** changed).
  - **Significance level ( $\alpha$ ):** A cutoff (like 5%) to decide if the result is rare or not.
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# Steps in Hypothesis Testing

## 1. State the Hypotheses

- **Null Hypothesis ( $H_0$ ):** The canteen is correct. Students eat 2 apples a day.
- **Alternative Hypothesis ( $H_1$ ):** The canteen is wrong. Students eat **less** (or more) than 2 apples a day.

## 2. Collect Data

Ask 30 students how many apples they eat per day and find the average.

## 3. Analyze the Data

Use math (like calculating the mean and standard deviation) to check if the new average is **significantly different** from 2.

## 4. Make a Decision

- If your results show a big enough difference, you can **reject the null hypothesis**.
  - If not, you **fail to reject it** — meaning the canteen's claim might still be true.
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# Hypothesis Testing: Significance Value ( $\alpha$ ) and Confidence Interval (CI)

## 1. Significance Value ( $\alpha$ )

The **significance value** ( $\alpha$ ) is the **threshold probability** we set **before hypothesis testing**.

It determines **when to reject the null hypothesis ( $H_0$ )**.

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### 1.1 What is $\alpha$ ?

$\alpha = 0.05$  (5%),  $\alpha = 0.01$  (1%),  $\alpha = 0.10$  (10%)

**Formula:**

$$\alpha = 1 - \text{Confidence Level}$$

**Example:**

- Confidence level = **95%**

$$\alpha = 1 - 0.95 = 0.05$$

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# P-value

## P-Value

- The **p-value** tells us **how likely our sample result (or something more extreme) is, if the null hypothesis ( $H_0$ ) were true.**
  - **Small p-value** → **evidence against  $H_0$ .**
  - Rule of thumb:
    - If  **$p < 0.05$**  → Reject  $H_0$  (significant).
    - If  **$p \geq 0.05$**  → Fail to reject  $H_0$  (not significant).
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# P-value: One-Tailed vs Two-Tailed Tests

## 1 One-Tailed Test

- Hypothesis checks only **one direction** (greater than or less than).
  - Example: Test if students eat **more than 1 apple/day** ( $H_1: \mu > 1$ ).
  - The p-value is the **area in one tail** (right side for  $\mu >$ , left side for  $\mu <$ ).
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## 2 Two-Tailed Test

- Hypothesis checks for **any difference** (not equal).
  - Example: Test if students eat **different from 1 apple/day** ( $H_1: \mu \neq 1$ ).
  - The p-value is the **area in both tails** (extreme low and extreme high).
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✓ Rule:

- **One-tailed** → Use when you care about only one direction.
- **Two-tailed** → Default choice when testing for “difference.”

# Step-by-Step Process

## 1. Set up Null and Alternate Hypotheses

- **H<sub>0</sub> (Null Hypothesis):** The statement we test (no change/no effect)
- **H<sub>1</sub> (Alternative Hypothesis):** What we want to prove (there is a change/effect)

## 2. Decide the Significance Level ( $\alpha$ )

- Common choices: 0.05 (5%) or 0.01 (1%)

## 3. Select the Right Test

- Use **Z-test** if population standard deviation is known
- Use **T-test** if population standard deviation is unknown and sample size  $< 30$

## 4. Calculate the Test Statistic (Z or T score)

- Formula for Z-score:

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

- Formula for T-score:

$$T = \frac{\bar{X} - \mu}{s / \sqrt{n}}$$

## 5. Find the p-value

## 6. Compare p-value and $\alpha$

- If **p <  $\alpha$** , reject the null hypothesis
  - If **p >  $\alpha$** , accept the null hypothesis
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|  Step |  Description                                                                                               |  Explanation / Example                                 |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1 State the claim                                                                    | Machine makes bolts with mean diameter <b>5 mm</b>                                                                                                                                         | Target ( $\mu_0$ ) = <b>5 mm</b>                                                                                                         |
| 2 Define hypotheses                                                                  | <ul style="list-style-type: none"> <li>- Null (<math>H_0</math>): <math>\mu = 5 \text{ mm}</math></li> <li>- Alternative (<math>H_1</math>): <math>\mu &gt; 5 \text{ mm}</math></li> </ul> | We test if the mean <b>has increased</b>                                                                                                 |
| 3 Collect sample data                                                                | Measure some sample bolts (e.g., 10 pieces)                                                                                                                                                | Example: 5.1, 5.0, 5.2, 5.3 ...                                                                                                          |
| 4 Calculate sample mean ( $\bar{x}$ )                                                | Find the average of sample diameters                                                                                                                                                       | Suppose $\bar{x} = \mathbf{5.15 \text{ mm}}$                                                                                             |
| 5 Perform test                                                                       | Use <b>t-test</b> or <b>z-test</b>                                                                                                                                                         | Formula:<br>$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$                                                                                     |
| 6 Set significance level ( $\alpha$ )                                                | Choose <b><math>\alpha = 0.05</math></b>                                                                                                                                                   | 5% chance of error                                                                                                                       |
| 7 Compare & conclude                                                                 | If <b>p-value</b> < <b><math>\alpha</math></b> , reject $H_0$                                                                                                                              | Example: $p = 0.02 \rightarrow$ Mean has increased  |

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## Example Scenario

A machine produces **bolts** with a target diameter of **5 mm**.  
We want to check if the **average diameter** has **increased**.

**Given:**

- Population standard deviation,  $\sigma = 0.4$  mm
  - Sample size,  $n = 25$
  - Sample mean,  $\bar{x} = 5.3$  mm
  - Significance level,  $\alpha = 0.05$
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## Step 1 — Define Hypotheses

**Case A: Right-Tailed Test** (Check if diameter has increased)

$H_0 : \mu = 5$  (machine is producing correct size)

$H_1 : \mu > 5$  (machine produces larger bolts)

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## Step 2 — P-value Approach

**Formula:**

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

**Calculation:**

$$Z = \frac{5.3 - 5}{0.4 / \sqrt{25}} = \frac{0.3}{0.08} = 3.75$$

**Find P-value**

Using  $Z = 3.75$  and right-tailed test:

$$p = P(Z > 3.75) \approx 0.00009$$

**Decision Rule:**

- If  $p < \alpha \rightarrow$  **Reject**  $(H_0)$
- Here,  $p = 0.00009 < 0.05$  ✓

**Conclusion:**

The bolts are **significantly larger** than 5 mm.

## ▼ 1. Independent t-test Example (Unpaired Groups)

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### Scenario

We want to compare **marks of boys and girls** in a math test.

| Group | Scores               |
|-------|----------------------|
| Boys  | [55, 60, 65, 70, 75] |
| Girls | [65, 70, 72, 68, 74] |

### Objective

Check if there is a **significant difference** between the **average marks** of boys and girls.

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|  Step |  Description                                                                             |  Explanation / Example |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1 State the claim                                                                     | Compare mean marks of <b>boys</b> and <b>girls</b>                                                                                                                        | Two independent groups                                                                                    |
| 2 Define hypotheses                                                                   | <ul style="list-style-type: none"> <li>- <math>H_0: \mu_1 = \mu_2</math> (means are equal)</li> <li>- <math>H_1: \mu_1 \neq \mu_2</math> (means are different)</li> </ul> | We test if there is any difference                                                                        |
| 3 Collect data                                                                        | Boys: [55, 60, 65, 70, 75]<br>Girls: [65, 70, 72, 68, 74]                                                                                                                 | Two separate samples                                                                                      |
| 4 Calculate sample means                                                              | Boys avg = <b>65.0</b><br>Girls avg = <b>69.8</b>                                                                                                                         | Girls have slightly higher marks                                                                          |
| 5 Perform t-test                                                                      | Use <b>independent t-test formula</b> to compare means                                                                                                                    | $t = -2.23$                                                                                               |
| 6 Set significance level ( $\alpha$ )                                                 | $\alpha = 0.05$                                                                                                                                                           | 5% chance of being wrong                                                                                  |
| 7 Compute p-value                                                                     | $p = 0.056$                                                                                                                                                               | Compare this with $\alpha$                                                                                |
| 8 Draw conclusion                                                                     | Since $p (0.056) > \alpha (0.05) \rightarrow$ <b>Fail to Reject <math>H_0</math></b>                                                                                      | There is <b>no significant difference</b> between boys' and girls' marks                                  |

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# Independent Samples t-test

## What it does

- Compares the **means** of two *independent groups*.
  - Helps to check if the difference in means is **significant** or just due to chance.
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## 1. Independent t-test Example (Unpaired Groups)

### Scenario:

You want to compare **marks of boys and girls** in a math test.

- Group 1 (Boys): [55, 60, 65, 70, 75]
  - Group 2 (Girls): [65, 70, 72, 68, 74]
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### Step-by-Step:

#### 1 Hypotheses:

- $H_0: \mu_1 = \mu_2$  (No difference in scores)
- $H_1: \mu_1 \neq \mu_2$  (There is a difference)

#### 2 Calculate sample means:

- Mean (Boys) = 65
- Mean (Girls) = 69.8

#### 3 Use independent t-test formula or calculator:

- Test statistic  $t \approx -2.02$
- Degrees of freedom  $\approx 8$

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- Critical t-value ( $\alpha = 0.05$ , two-tailed)  $\approx \pm 2.306$

#### 4 Decision:

- $|t| = 2.02 < 2.306 \rightarrow \text{Fail to reject } H_0$

 **Conclusion:** No significant difference between boys' and girls' scores.

## 2. Dependent t-test Example (Paired Groups)

### Scenario:

You want to test if a **coaching class improved scores**. You take scores of **same students** before and after the class.

- **Before:** [50, 55, 52, 60, 58]
- **After:** [55, 60, 58, 65, 64]

### Step-by-Step:

#### 1 Hypotheses:

- $H_0: \mu_d = 0$  (No improvement)
- $H_1: \mu_d > 0$  (Improved after class)

#### 2 Find differences:

| Student | After | Before | Difference (d) |
|---------|-------|--------|----------------|
| 1       | 55    | 50     | 5              |
| 2       | 60    | 55     | 5              |
| 3       | 58    | 52     | 6              |
| 4       | 65    | 60     | 5              |
| 5       | 64    | 58     | 6              |

- Mean of d = 5.4
- SD of d  $\approx 0.55$
- n = 5

#### 3 t-statistic:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} = \frac{5.4}{0.55 / \sqrt{5}} \approx 21.97$$

- Critical t ( $\alpha = 0.05$ , df = 4, one-tailed)  $\approx 2.132$

#### 4 Decision:

- $21.97 > 2.132 \rightarrow \checkmark$  **Reject  $H_0$**

**Conclusion:** The class **significantly improved** scores.

|  Step |  Description                                                                            |  Explanation / Example                                    |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1 State the claim                                                                     | Check if coaching <b>increased</b> scores                                                                                                                                | Same students before & after                                                                                                                 |
| 2 Define hypotheses                                                                   | <ul style="list-style-type: none"> <li>- <math>H_0: \mu_d = 0</math> (mean difference = 0)</li> <li>- <math>H_1: \mu_d &gt; 0</math> (mean difference &gt; 0)</li> </ul> | One-tailed test (we expect increase)                                                                                                         |
| 3 Collect paired data                                                                 | Before and After lists (same students)                                                                                                                                   | Before: [50,55,52,60,58]<br>After: [55,60,58,65,64]                                                                                          |
| 4 Compute differences                                                                 | $d = \text{After} - \text{Before}$                                                                                                                                       | $d = [5, 5, 6, 5, 6]$                                                                                                                        |
| 5 Calculate mean of differences ( $\bar{d}$ )                                         | $\bar{d} = (5+5+6+5+6)/5 = 5.4$                                                                                                                                          | Average improvement per student                                                                                                              |
| 6 Calculate sd of differences ( $s_d$ )                                               | $s_d \approx 0.548$                                                                                                                                                      | Sample standard deviation of d                                                                                                               |
| 7 Compute t-statistic                                                                 | $t = \bar{d} / (s_d / \sqrt{n})$                                                                                                                                         | $t \approx 22.05$ ( $n = 5$ )                                                                                                                |
| 8 Set significance level ( $\alpha$ )                                                 | $\alpha = 0.05$                                                                                                                                                          | 5% risk of error                                                                                                                             |
| 9 Compute p-value                                                                     | $p < 0.0001$ (very small)                                                                                                                                                | Compare p with $\alpha$                                                                                                                      |
| 10 Draw conclusion                                                                    | Since $p < \alpha$ , <b>Reject <math>H_0</math></b>                                                                                                                      |  <b>Coaching improved scores</b> (significant increase) |

# ANOVA (Analysis of Variance)

## Full Form

ANOVA = Analysis of Variance

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## Introduction

- A statistical method to compare the **means of 3 or more groups**.
- Checks whether differences in sample means are **statistically significant** or just due to random variation.

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## One-Way ANOVA Example (Manual Calculation)

### Objective:

Test if there is a significant difference in average scores among **three different classes** of students.

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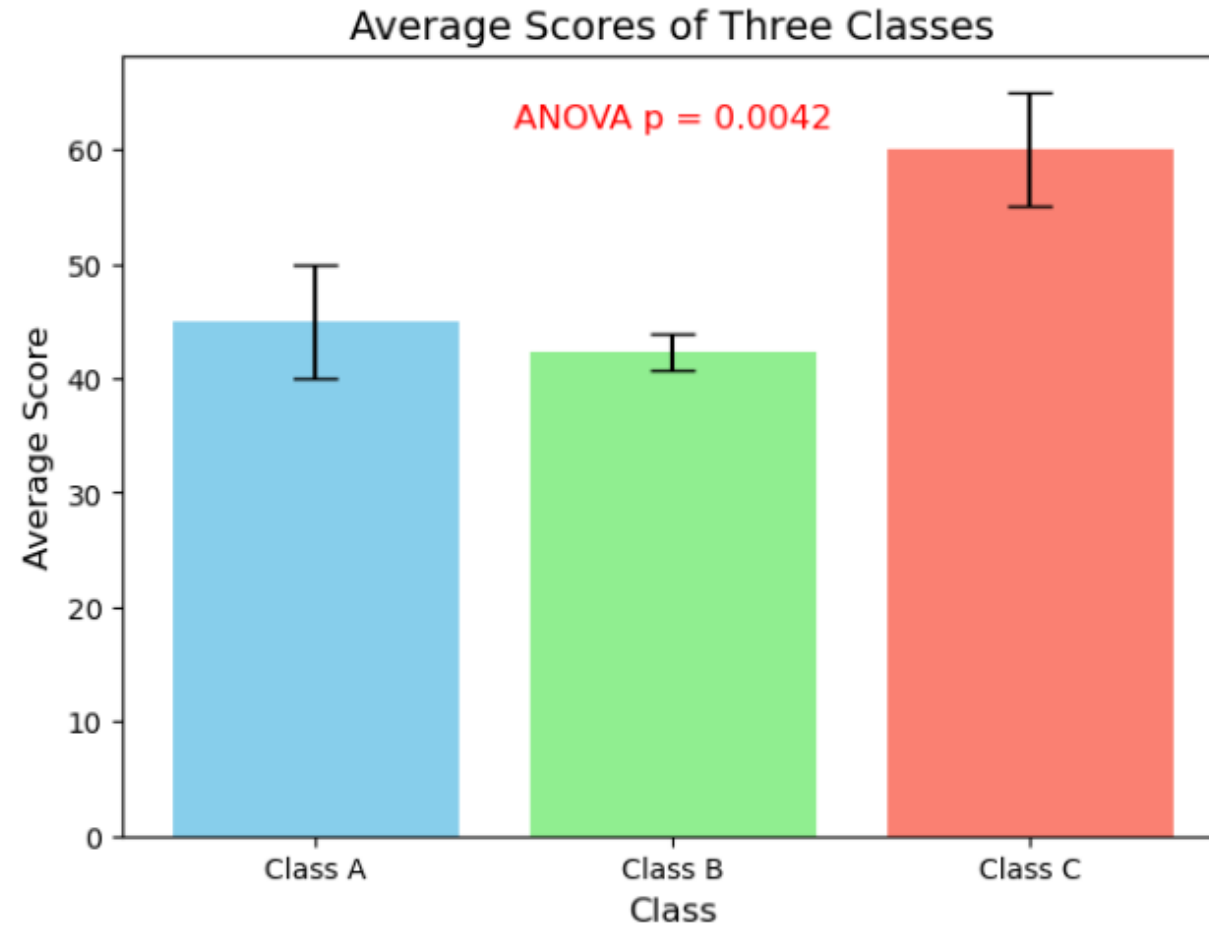
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### Data Table:

| Class A | Class B | Class C |
|---------|---------|---------|
| 40      | 42      | 55      |
| 45      | 41      | 60      |
| 50      | 44      | 65      |

- **k** = 3 groups
  - **n** = 3 observations per group
  - **N** = 9 total observations
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F-statistic = 15.60, p-value = 0.0042



ince p-value = 0.0009 < 0.05, we reject  $H_0$ .

That means: Not all class averages are equal. At least one class differs significantly.



# Chi-Square Test ( $\chi^2$ Test)



## What is a Chi-Square Test?

The **Chi-Square ( $\chi^2$ ) Test** is a statistical method used to:

- Compare **observed** values with **expected** values.
- Check if there is a **significant association** between two categorical variables.



## Example: Chi-Square Test of Independence



### Situation:

We want to know whether **gender** and **preference for a subject** are related.

## Step 1 — Hypotheses

- $H_0$ : Gender and subject choice are **independent** (no relation).
- $H_1$ : Gender and subject choice are **not independent** (there is a relation)



### Data (Observed):

|       | Science | Arts | Total |
|-------|---------|------|-------|
| Boys  | 30      | 10   | 40    |
| Girls | 10      | 30   | 40    |
| Total | 40      | 40   | 80    |



## Step 1: Find Expected Values

$$E_{ij} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

### Example:

Expected value for Boys–Science:

$$E = \frac{40 \times 40}{80} = 20$$

|       | Science (E) | Arts (E) |
|-------|-------------|----------|
| Boys  | 20          | 20       |
| Girls | 20          | 20       |



## Step 2: Apply Chi-Square Formula

Formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

### Calculations:

- For Boys–Science:  $(\frac{(30 - 20)^2}{20} = 5)$
- For Boys–Arts:  $(\frac{(10 - 20)^2}{20} = 5)$
- For Girls–Science:  $(\frac{(10 - 20)^2}{20} = 5)$
- For Girls–Arts:  $(\frac{(30 - 20)^2}{20} = 5)$

$$\text{Total } \chi^2 = 5 + 5 + 5 + 5 = 20$$



## Step 3: Degrees of Freedom

$$df = (r - 1)(c - 1) = (2 - 1)(2 - 1) = 1$$

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### ✓ Step 4: Compare with $\chi^2$ Table

- At  $\alpha = 0.05$  and  $df = 1$ , critical value  $\approx 3.84$
- Our  $\chi^2 = 20$ , which is **greater** than 3.84

✓ **Conclusion:** Reject  $H_0$

There is a significant relationship between gender and subject preference.

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|  Step                      |  Description                                                                                                                             |  Explanation / Example                               |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>1 State the claim</b>                                                                                   | We test if <b>gender</b> and <b>subject preference</b> are <b>independent</b>                                                                                                                                            | Boys & girls choose subjects differently?                                                                                              |
| <b>2 Define hypotheses</b>                                                                                 | <ul style="list-style-type: none"> <li>- <b>H<sub>0</sub></b>: Gender and subject preference are <b>independent</b></li> <li>- <b>H<sub>1</sub></b>: Gender and subject preference are <b>not independent</b></li> </ul> | Two-way relationship test                                                                                                              |
| <b>3 Write observed frequencies (O)</b>                                                                    | Use the given data table                                                                                                                                                                                                 | e.g., $O_{11} = 30$ , $O_{12} = 10$ , etc.                                                                                             |
| <b>4 Compute expected frequencies (E)</b>                                                                  | $E = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$                                                                                                                                                 | For Boys–Science: $(40 \times 40) / 80 = 20$                                                                                           |
| <b>5 Calculate all expected (E)</b>                                                                        | Boys–Science = 20<br>Boys–Arts = 20<br>Girls–Science = 20<br>Girls–Arts = 20                                                                                                                                             | Expected table: all = 20                                                                                                               |
| <b>6 Apply Chi-Square formula</b>                                                                          | $\chi^2 = \sum ((O - E)^2 / E)$                                                                                                                                                                                          | $= ((30 - 20)^2 / 20) + ((10 - 20)^2 / 20) + ((10 - 20)^2 / 20) + ((30 - 20)^2 / 20)$                                                  |
| <b>7 Compute <math>\chi^2</math> value</b>                                                                 | $\chi^2 = 10 + 5 + 5 + 10 = 30$                                                                                                                                                                                          | Calculated test statistic                                                                                                              |
| <b>8 Find degrees of freedom (df)</b>                                                                      | $df = (r - 1)(c - 1) = (2 - 1)(2 - 1) = 1$                                                                                                                                                                               | One degree of freedom                                                                                                                  |
| <b>9 Set significance level (<math>\alpha</math>)</b>                                                      | $\alpha = 0.05$                                                                                                                                                                                                          | 5% risk of error                                                                                                                       |
| <b>10 Compare with critical value or p-value</b>                                                           | For $df=1$ , $\chi^2_{0.05} = 3.84$                                                                                                                                                                                      | $\chi^2 = 30 > 3.84$                                                                                                                   |
|  <b>Draw conclusion</b> | Since $\chi^2 > 3.84 \rightarrow$ <b>Reject H<sub>0</sub></b>                                                                                                                                                            |  <b>Gender and subject preference are related</b> |

